Conversion of Camphene to Camphor

Day 3: Conversion of Borneol to Camphor

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Chem 2081 007

Introduction

In this experiment, borneol was oxidized to form camphor using acetic acid and acetone. Camphor was then extracted, dried, and weighed.

Materials and Methods

First, 0.3 g of borneol was added to 0.3 mL of glacial acetic acid in a 10 mL long-neck round-bottom flask. To this mixture was added 1 mL of acetone and a boiling chip. An open reflux setup was constructed according to Figure 1.18 on page 20 of CHEM 2081 Organic Chemistry 2 Lab Manual (Keller, 2023). The apparatus was placed in a hot water bath, and the temperature kept strictly between 40-50 °C. After all the borneol was dissolved, 6 mL of aqueous sodium hypochlorite was added dropwise, with swirling, to the mixture through the open top of the apparatus, over 35 minutes. After the last addition, the reaction was allowed to continue for an additional 15 minutes. The mixture was then removed from the water bath and allowed to cool to room temperature. It was subsequently extracted twice with 2 mL of dichloromethane (totaling 4 mL) using a centrifuge tube. The remaining solution was then neutralized with 2 mL saturated sodium bicarbonate and extracted again. This solution was then washed sequentially with 2 mL of 5% sodium bisulfite and 2 mL of water, then extracted one final time. The solution was then dried with anhydrous sodium sulfate, decanted, and then evaporated on a watch glass over the hot water bath, still not exceeding 50 °C. The watch glass and the watch glass with product were weighed. The reflux setup was disassembled, the waste properly disposed of, and the glassware was washed and left to dry.

Results

Amount of Bleach (mL)	6
Amount of Borneol (g)	0.3

Weight of Dry Watch Glass (g)	37.254
Weight of Watch Glass and Camphor (g)	37.448

Table 1. Amounts of reagents and weights of watch glass, camphor covered watch glass, and camphor

Calculations

$$0.3\ g * \frac{1\ mol}{154.25\ g} = 0.0019\ mol\ borneol$$

$$6\ mL * \frac{1\ g}{1\ mL} = 6\ grams\ of\ bleach$$

$$6\ g * \frac{1\ mol}{74.442\ g} = 0.08\ mol\ bleach$$

Therefore, borneol is the limiting reactant.

$$C_{10}H_{18}O + ClO^- + [H^+] \rightarrow C_{10}H_{16}O + H_3O^+ + Cl^-$$

$$0.0019\ mol\ C_{10}H_{18}O * \frac{1\ mol\ C_{10}H_{16}O}{1\ mol\ C_{10}H_{18}O} = 0.0019\ mol\ C_{10}H_{18}O$$

$$0.0019\ mol * \frac{152.23\ g}{1\ mol} = 0.296\ grams\ of\ camphor\ (theoretical)$$

$$37.448\ g - 37.254\ g = 0.194\ grams\ of\ camphor\ (experimental)$$

$$\frac{0.194\ g}{0.296\ g} * 100\% = 65.54\%\ yield$$

Discussion

In this experiment, borneol was oxidized to camphor with bleach and glacial acetic acid. First, a proton transfer occurs to make the alcohol a good leaving group. Then, hypochlorite attacks, pushing off the leaving group. This water molecule then takes the hydrogen attached to the carbon, forming a hydronium ion, camphor, and a chloride ion. This experiment had a 65.54% yield of camphor. Probable causes of this diminished yield include not allowing the

reaction to proceed long enough, inaccurate separation of the layers during liquid-liquid extraction, and allowing the water bath to exceed the recommended temperature. The most likely source of error was stopping the reaction before it had proceeded to completion. Since there is no visible way to determine how much of the camphor has formed, it is important to leave the reaction in the water bath after all of the bleach is added. One way to correct this error would be to leave the apparatus in the hot water bath for longer to ensure that all of the reagent reacted.

References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (AL): Auburn University.

Lab Performance

Exp 6 D3

Name: *Section:* *Room:* *Hood:*

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves.	○ ○ ○ ○ ○ (5pts)
Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks.	○ ○ ○ ○ ○ (5pts)
Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts).	○(1pt) ○(1pt) ○(1pt) ○ (1pt) ○ (5pts)
Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts).	○(1pt) ○ (5pts) ○ (5pts)
Working independently or equal partner participation during the entire lab.	○ ○ ○ ○ ○ (5pts)
Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace.	○ ○ ○ ○ ○ (5pts)
Maintain clean and organized personal space.	○ ○ ○ ○ ○ (5pts)

GTA notes:

GTA signature: _____ *Final Performance Grade:* _____

Exp. No. 6	Experiment/Subject Borneol to Camphor	Date 11/16	
Name Mathew Johnson	Lab Partner	Locker/ Desk No.	Course & Section No.

Procedure

1.) 10mL LNRB flask, add racemic .3g borneol product w/ .3mL glacial acetic acid. Dissolve the racemic mixture in 1mL acetone. Add BC.

2.) Assemble reflux w/ water bath temp around 40–50°C. If no dissolution, add 0.5mL acetone

3.) Add 6mL aq sodium hypochlorite (drop wise over 35 mins.) swirl after addition 15 mins after last addition.

4.) Let mix cool to room temp, extract camphor w/ 2 portions of methylene chloride (DCM) Use a centrifuge tube.

5.) Neutralize layers w/ 2mL sat. Sodium bicarbonate. keep adding till no CO_2 bubbles.

8.) Wash soln sequentially w/ 2mL 5% sodium bisulfite + 2mL H_2O.

7.) Dry soln over anhydrous sodium sulfate. Free rolling.

8.) Evaporate DCM over a water bath w/ fume hood exceeding 50°C. Weigh collected crystals + determine percent yield.

Observations

1.) Wt ~~0.0026~~ ~~10.0026~~
AA .3mL
.3g Borneol
1mL acetone

2.) ✓

3.) ✓ Go pretty slow... 1/2 mL over minute

4.) ✓ ρ DCM = 1.33 so product will be BOTTOM layer!

5.) ✓

~ spatula tip's worth per mL

Leave drying agent in centrifuge tube

w/product 37.498
w/o -37.259
total product.

Signature	Date	Witness/TA	Date